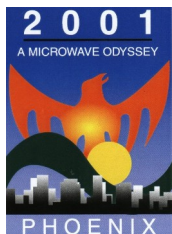


3G Transmitter Linearisation

An Alternative Approach

Dr. Richard G. Ranson
Director Wireless Research
Filtronic plc



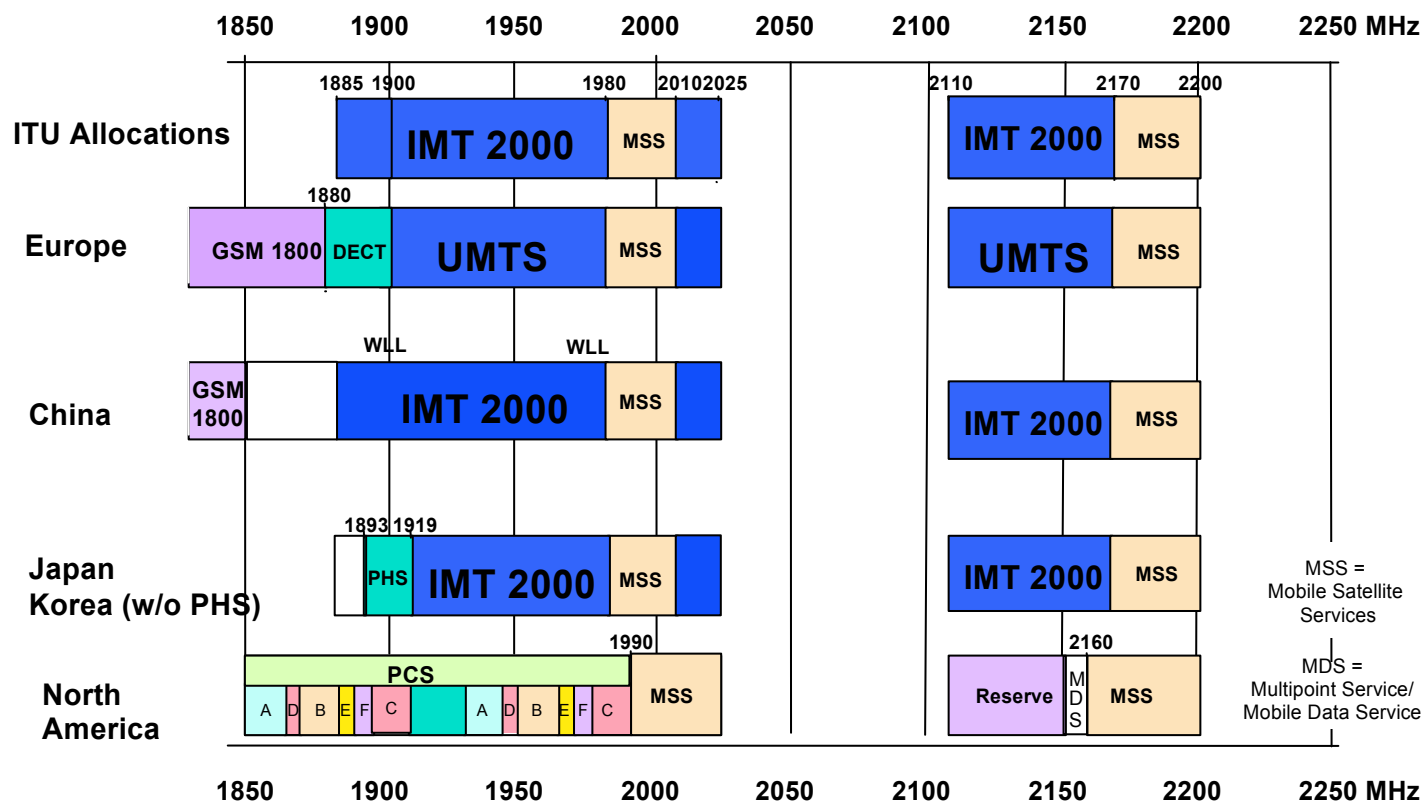
Contents

- Introduction
- Key Specification Issues
- Conventional View
- Alternative View
- Demonstration
- Conclusions

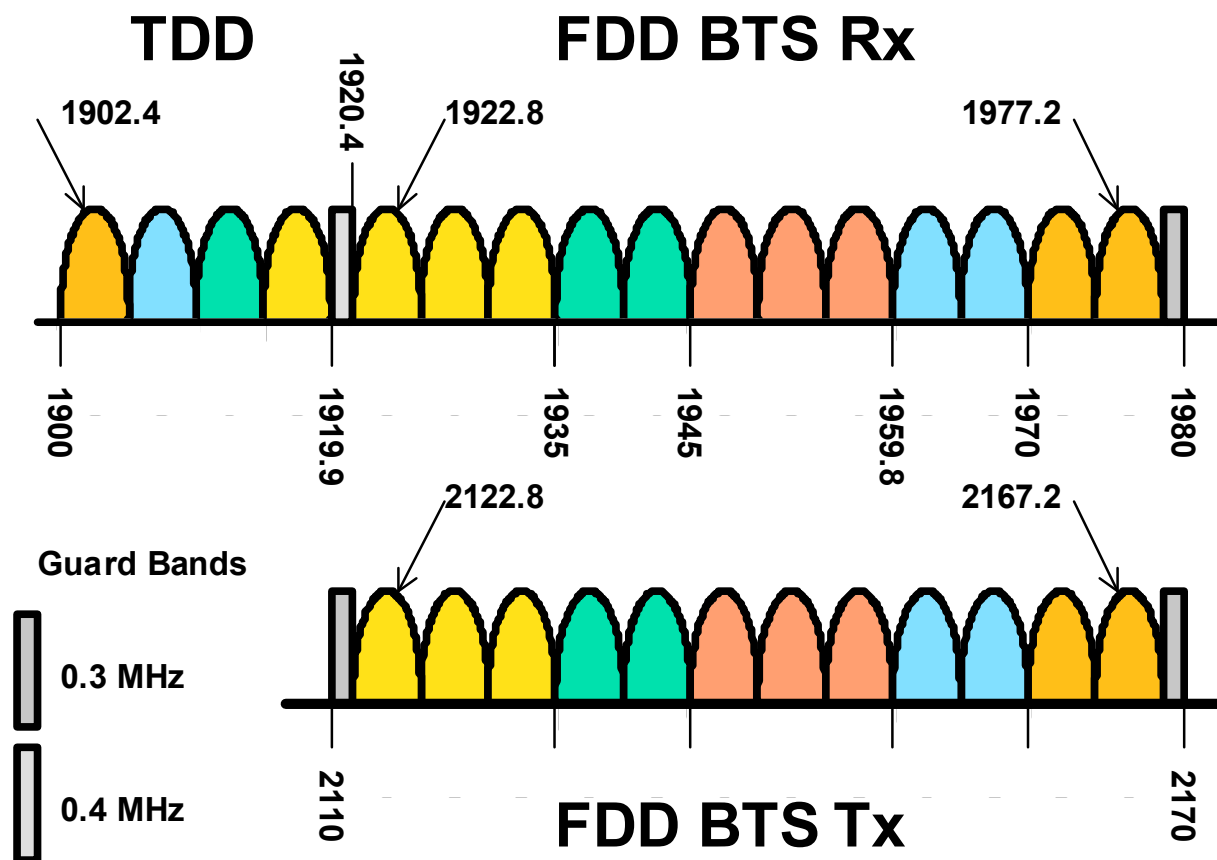
3G Systems

- CDMA 2000
 - Derivative of IS-95 (cdma-one)
 - 3, IS-95 signals in 3.75 MHz
- UTRA W-CDMA
 - 3GPP, single direct sequence signal
- Global agreement on frequ. allocation 1900 – 1980 & 2110 – 2170 MHz
 - Exception of North America, PCS band

WRC Frequency Allocations



Example UMTS Band Plan (UK)



UMTS Frequency Plan

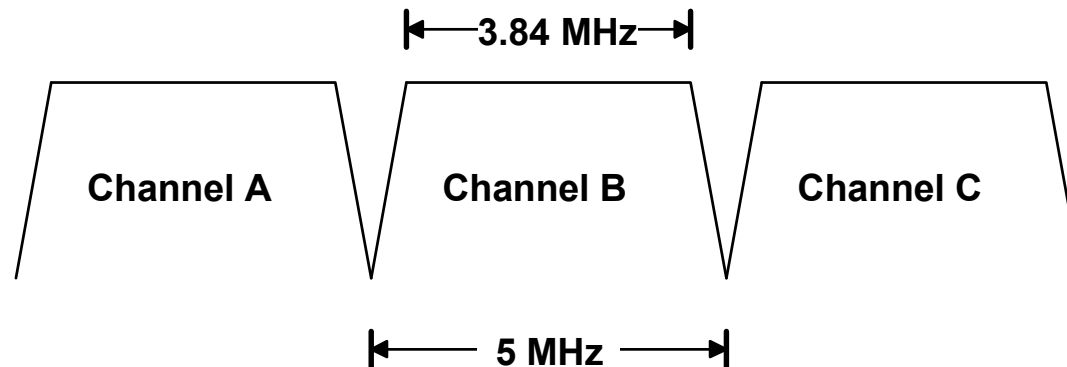
- FDD Allocations fall in 5 MHz segments
 - Only 12 segments in the band
 - Operators with 2 or 3 segments each
- 5 MHz channel allocation
 - Modest % bandwidth
 - <1.5 dB IL for $Q_u > 40,000$

Key Specification Points

- 3GPP Specification TS 25.104 v3.3
 - www.3gpp.org
- 6.6.1 Occupied bandwidth
- 6.6.2 Out of band emissions
- 6.8 Transmit modulation

Occupied Bandwidth

- 99% power within 3.84 MHz centred in a channel 5 MHz wide

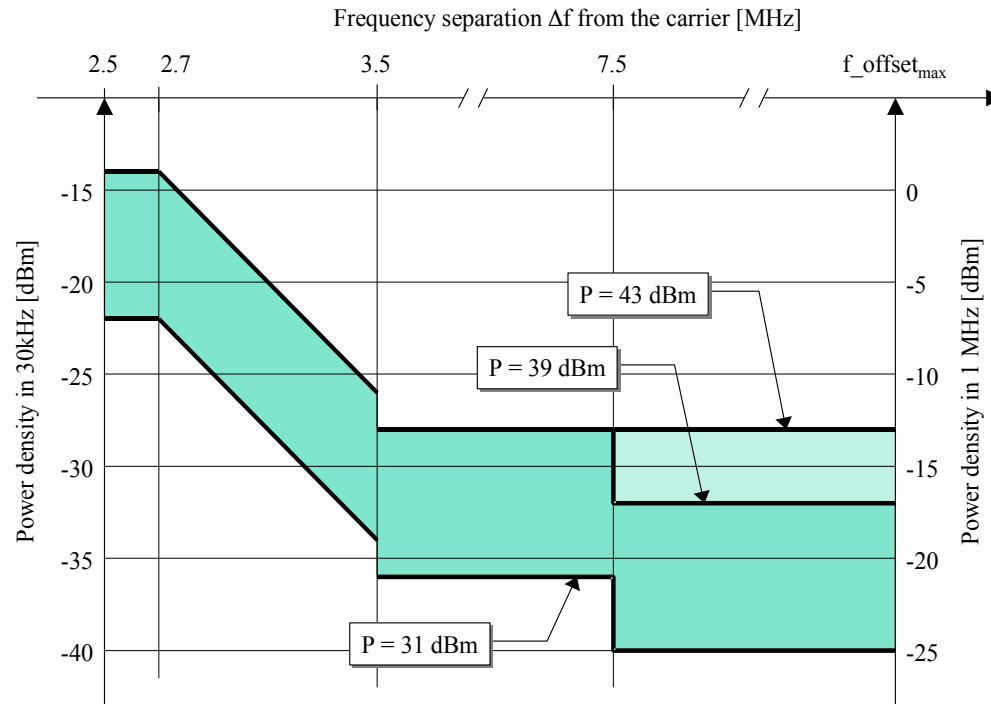


- Unlike IS-95 there is space between adjacent channels

Out of band emissions

- Spectrum Mask
 - dBc profile for adjacent channel and other out of band frequencies.
- Adjacent Channel Leakage Ratio
 - Same as ACPR, measured in 3.84 MHz
 - 5 MHz 45 dB
 - 10 MHz 50 dB

Spectrum Mask



Illustrative diagram of spectrum emission mask

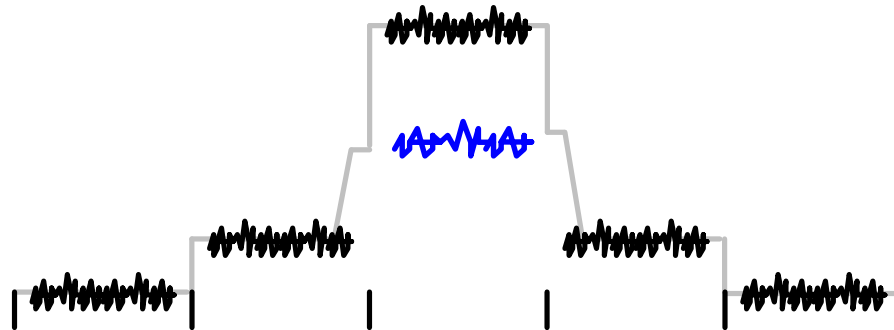
- Note that up to 4 MHz offset is specified in 30 kHz while beyond in 1MHz BW

Transmit modulation

■ Modulation Accuracy

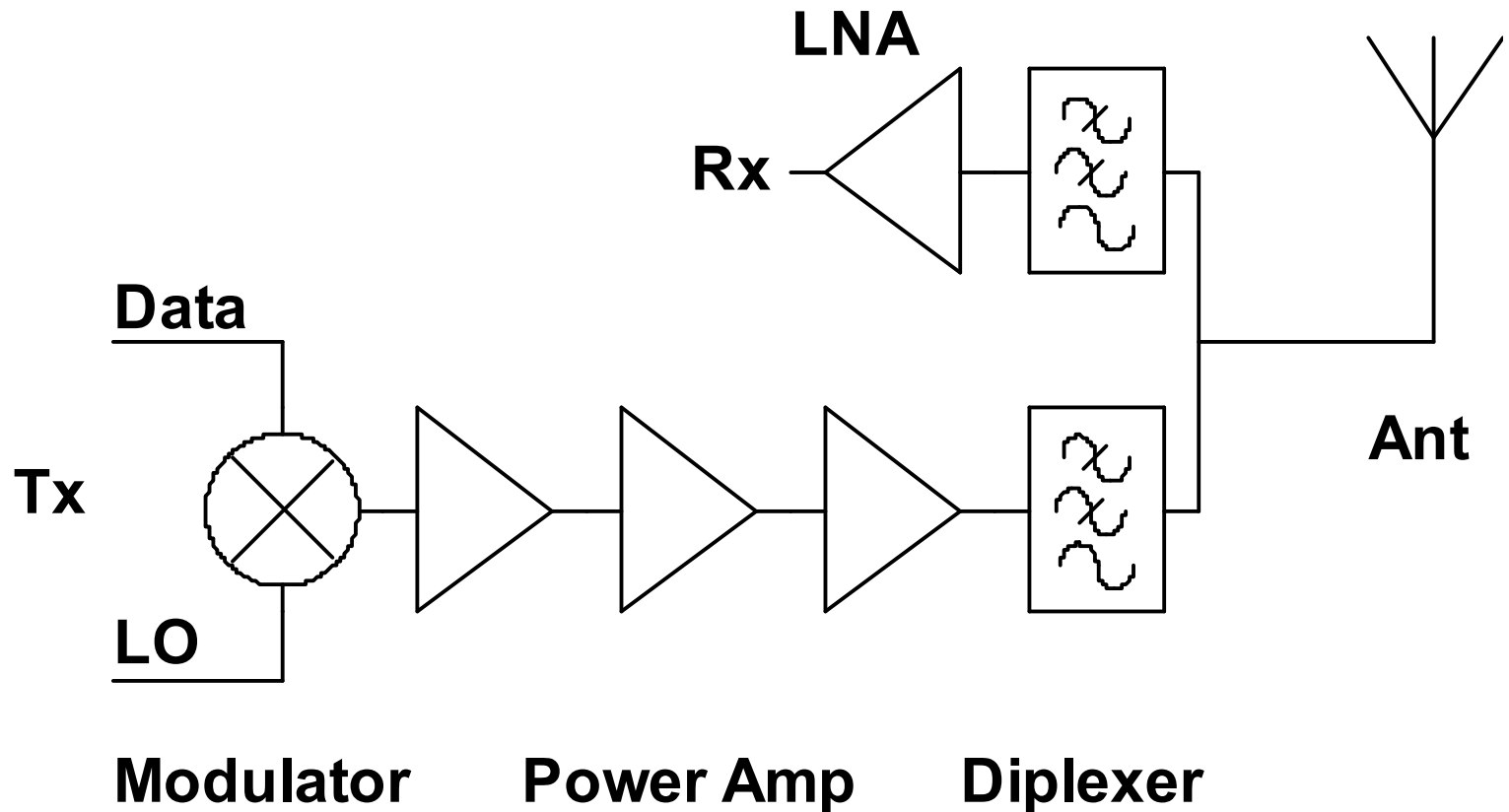
- Error Vector Magnitude
 - RMS EVM < 17.5%
- Code Domain Error
 - Pk CDE < -33 dB
- These are essentially measures of in band distortion

In band / Out of band IM



- Single 3.84 MHz signal in 5 MHz channel
- Spectrum mask for allowed emissions
- ACLR levels for each channel
- In band distortion allowance (EVM and CDE)

Basic System Block Diagram



Conventional View

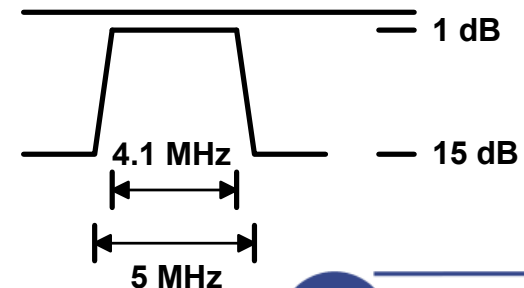
- Modulation Accuracy
 - Need a fantastic modulator
- Adjacent channel power
 - Need a super linear PA
- Occupied bandwidth
 - Duplexer separates Tx from Rx

Tx Spurious Approaches

- Very high performance linearised PA
 - Several techniques
 - Added complexity to get to the last 10 dB
 - Tough to get high efficiency
- High performance PA
 - Amplifier meets most of the spec
 - Eases complexity
 - Filter used to clean up the remainder

Tx Spurious – An Alternative

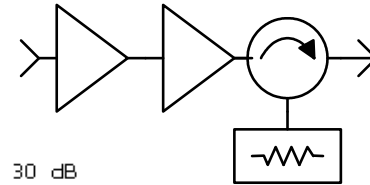
- There are distinct 5 MHz channels
- High Performance Filter
 - Much simpler amplifier with good efficiency
- Filter to clean up the spectrum
 - 1 dB more loss
 - Assume 3 : 1 IM growth
 - 5 dB more drive
 - Net 4 dB more output



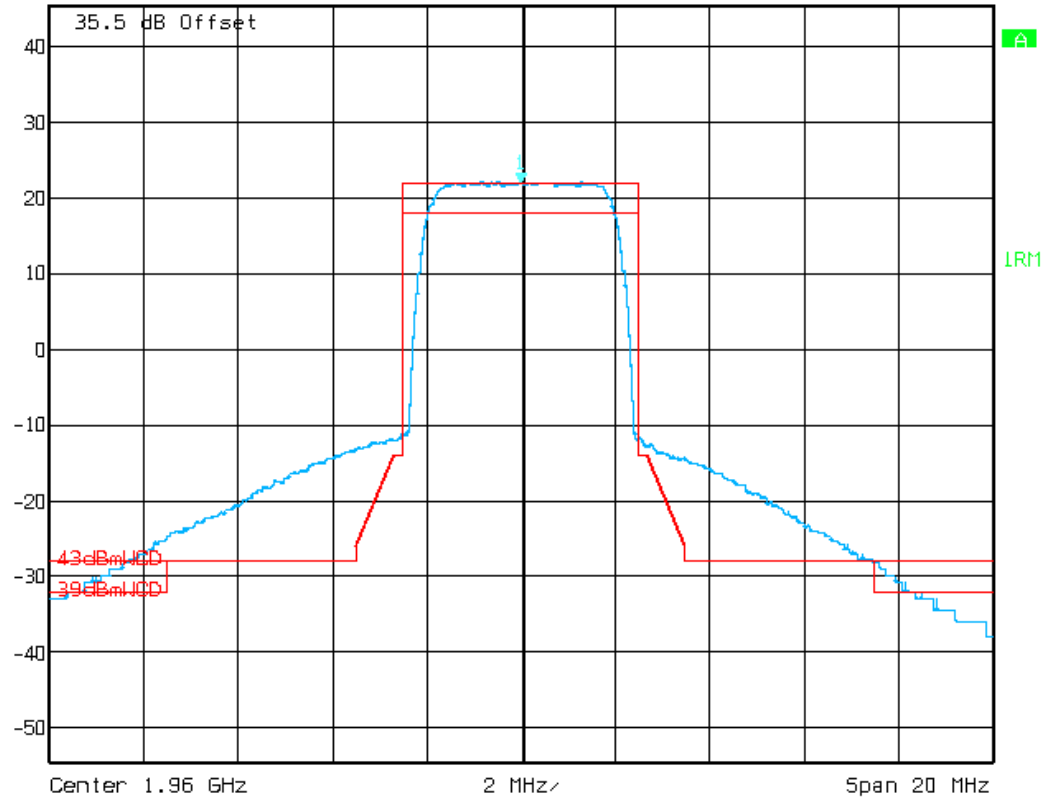
Efficiency and BTS Power

- GSM BTS 4 carriers, 3 sectors, 40 watts
 - $12 \times 40 = 480 \text{ W}$
 - $\eta = 45\%$, Total DC power 1.1 kW
- W-CDMA 2 carriers, 3 sectors, 30 watts
 - $6 \times 30 = 180 \text{ W}$,
 - $\eta = 8\%$, DC power 2.25kW
 - $\eta = 15\%$, 1.2 kW
 - $\eta = 20\%$, 0.9 kW

Output Spectrum

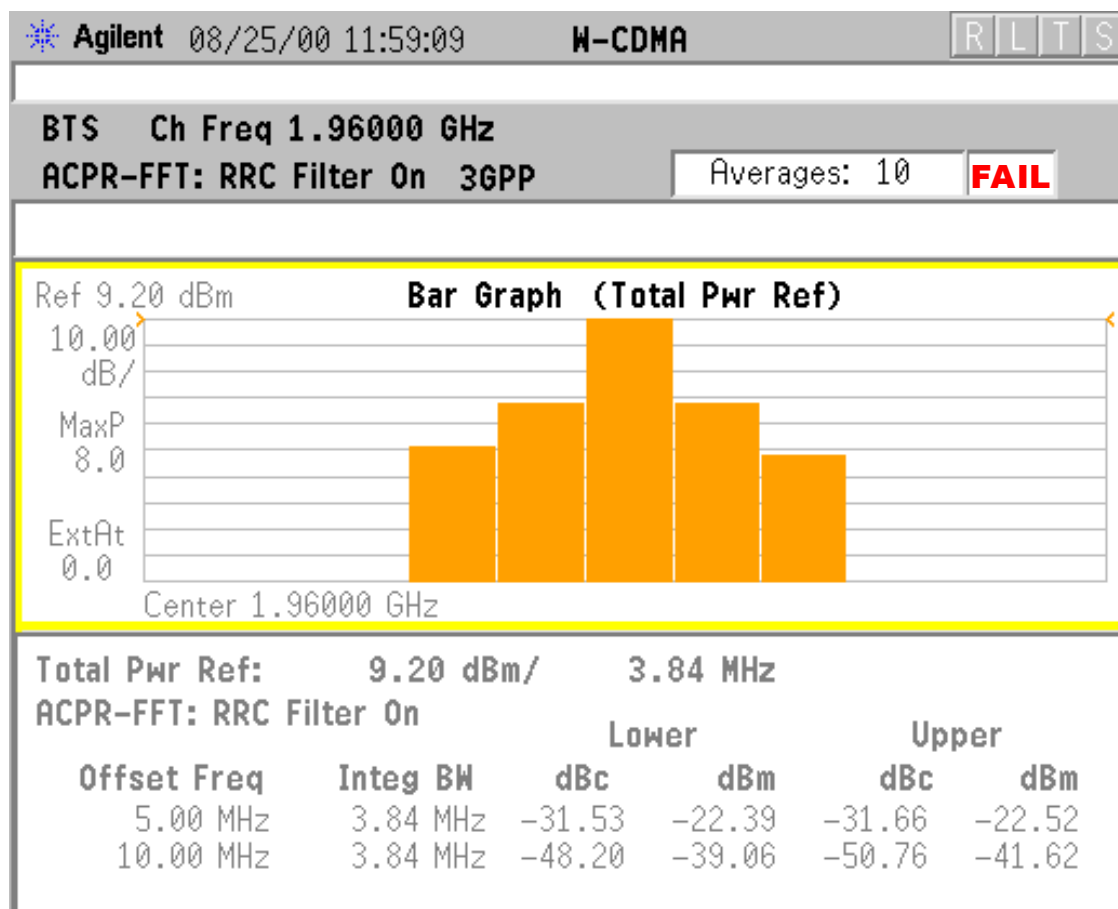
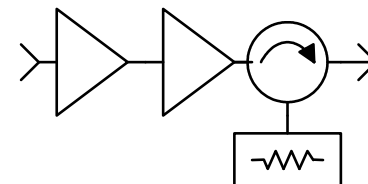


Ref Lvl 45.5 dBm
Marker 1 [T1] 21.91 dBm
1.95997996 GHz
RBW 30 kHz
VBW 300 kHz
SWT 10 s
RF Att 30 dB
Unit dBm

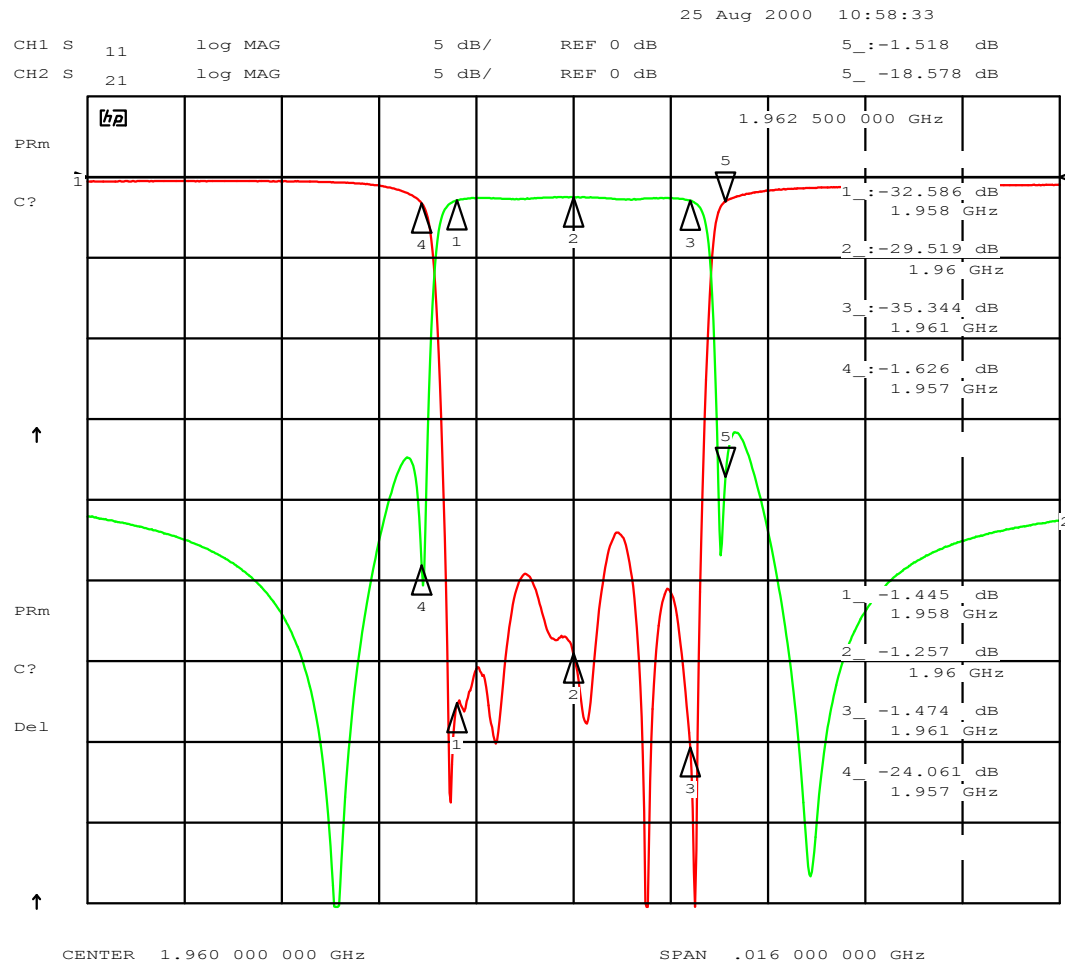


Date: 24.AUG.00 16:58:43

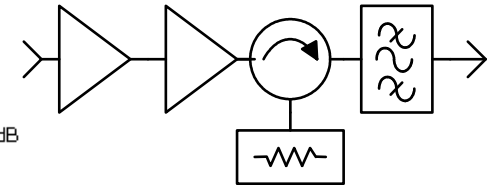
System Performance



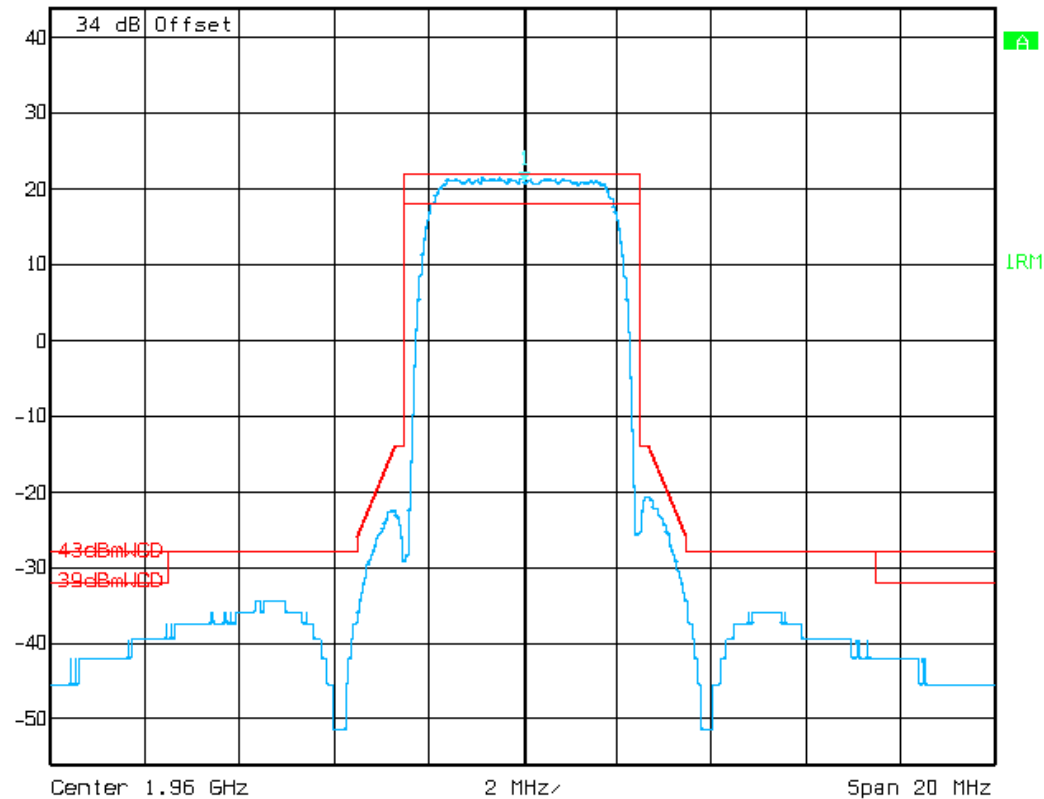
Filter Performance



Output Spectrum



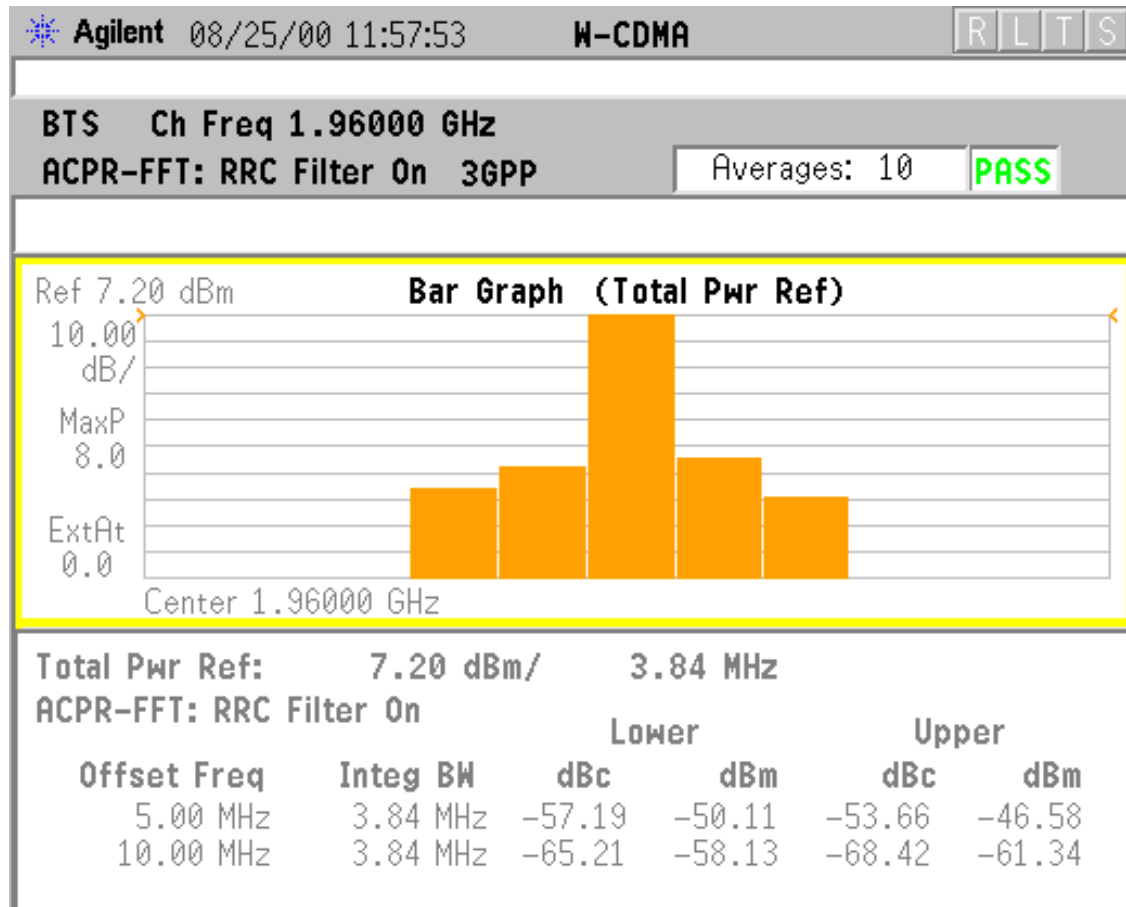
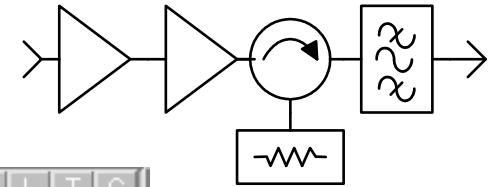
 Ref Lvl 44 dBm
 Marker 1 [T1] 21.09 dBm
 1.96000000 6Hz
 RBW 30 kHz RF Att 30 dB
 VBW 1 MHz
 SWT 10 s Unit dBm



Date: 25 AUG 00 11:23:57



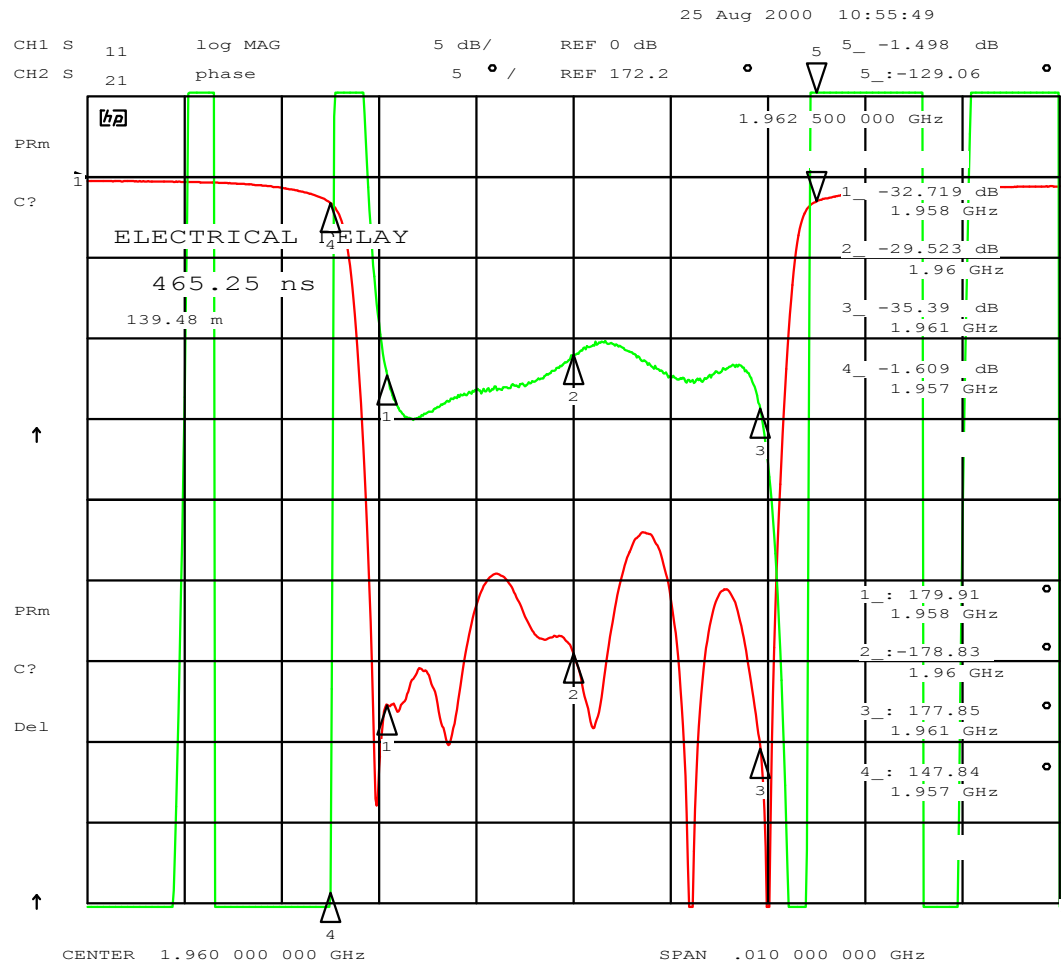
System Performance



Single Channel Tx Filter

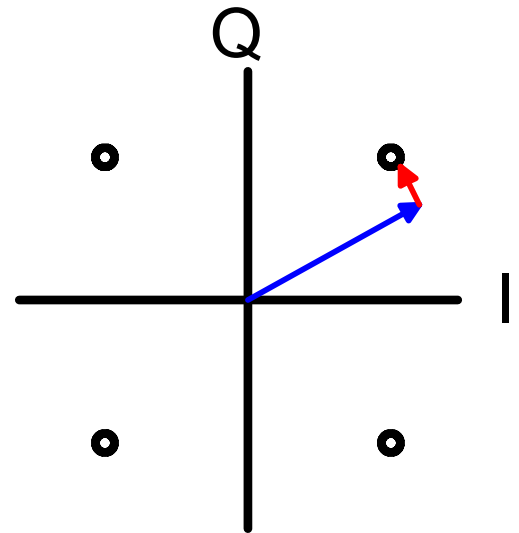
- Stopband Amplitude response
 - ACPR
 - Close in Spectral Mask
- Passband Amplitude and Phase
 - In band RMS EVM
 - Pk Code Domain Error

Filter Performance

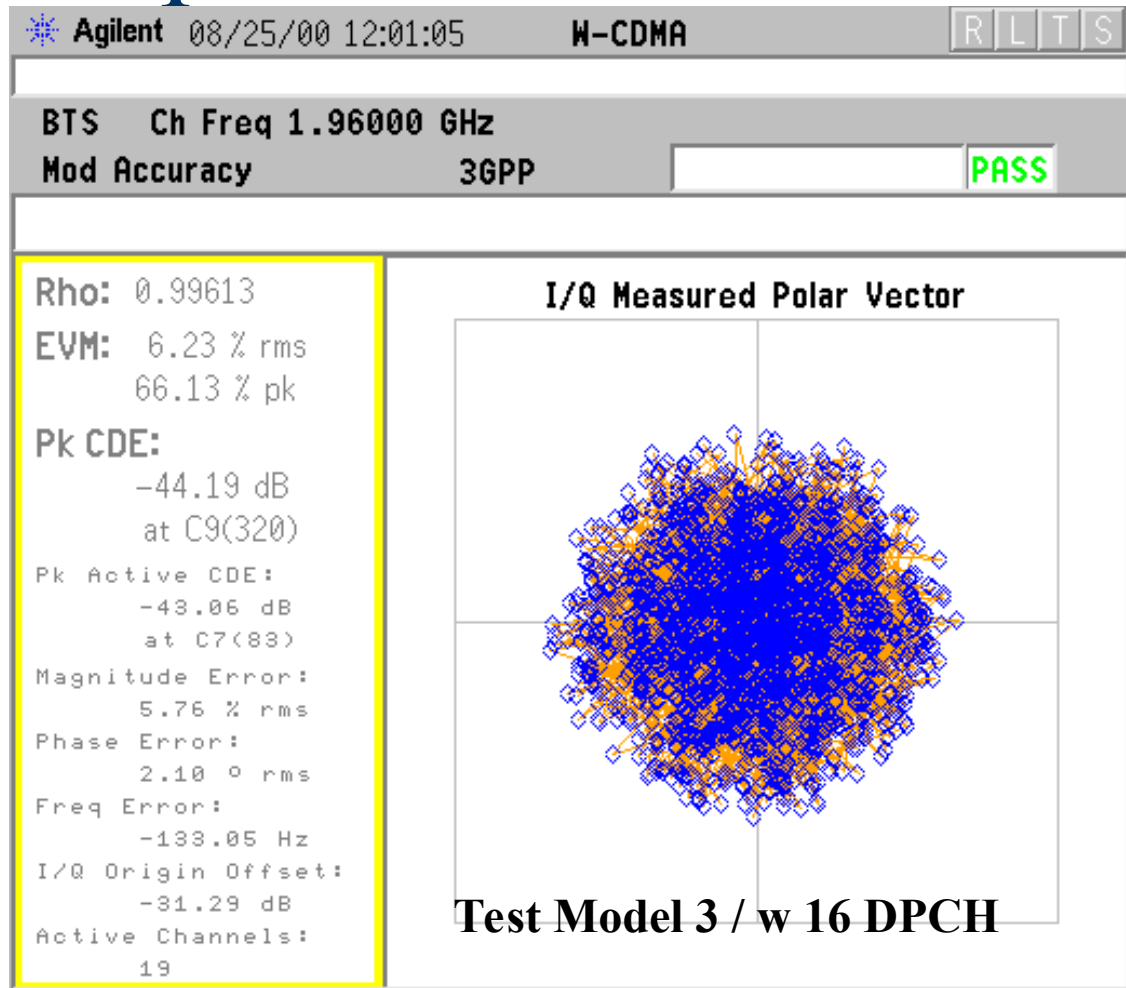
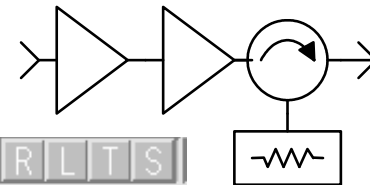


EVM Measurement

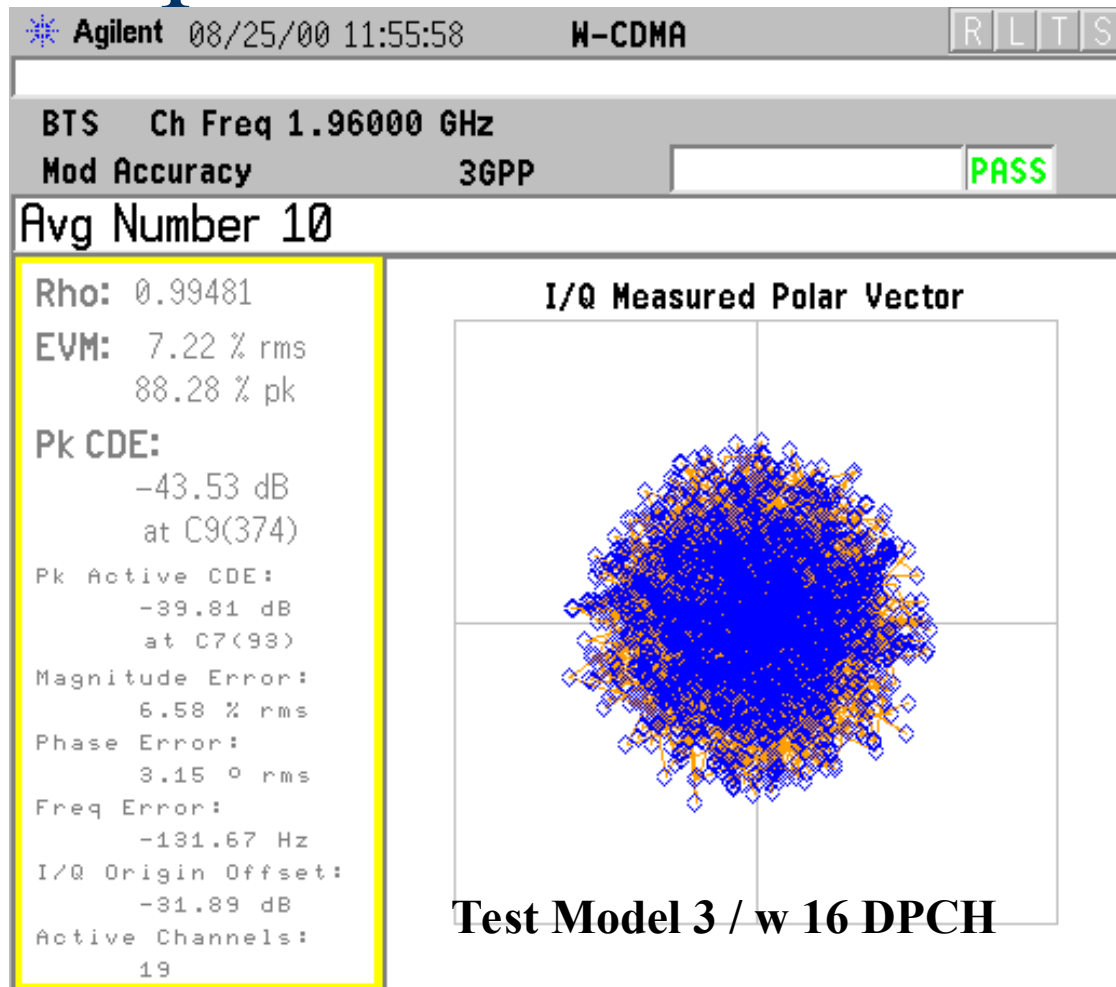
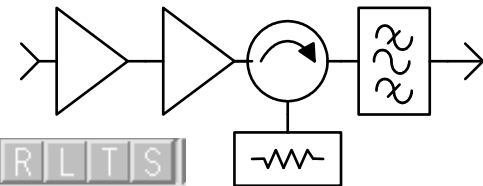
- Single QPSK signal
 - Error vector between actual signal and the reference points
 - Statistical average over many data points (RMS and Pk)
- Multiple signal
 - Composite EVM
 - Superposition of many codes
 - Same statistical averages



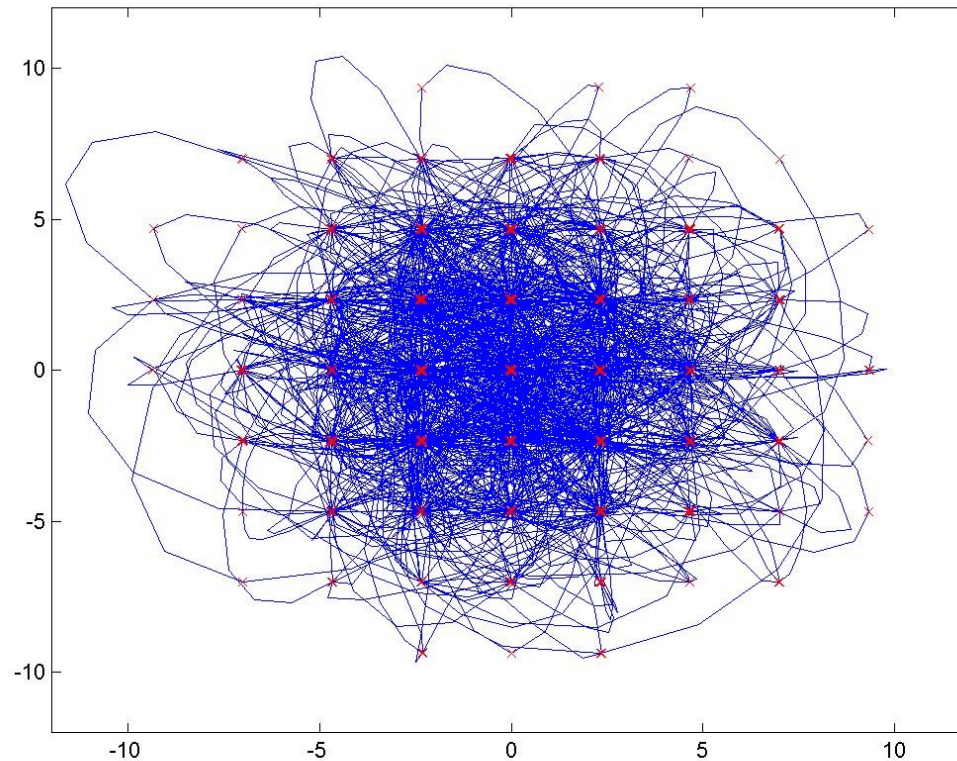
Composite EVM



Composite EVM

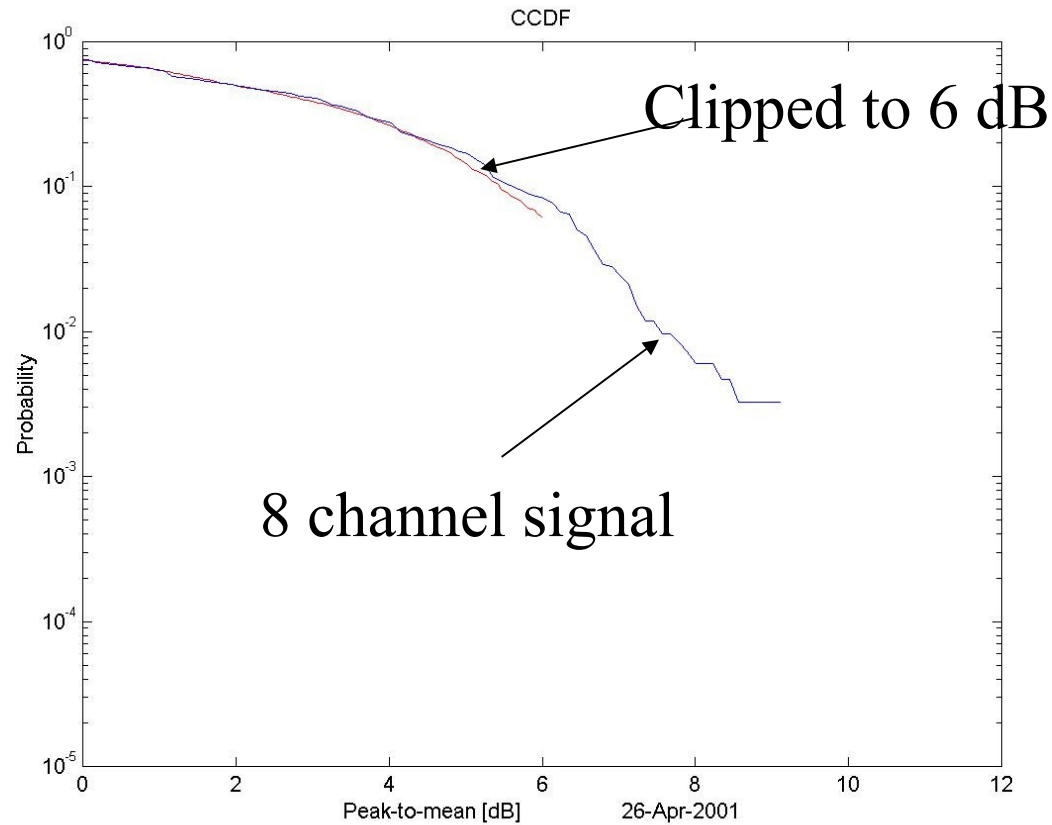


EVM Simulations

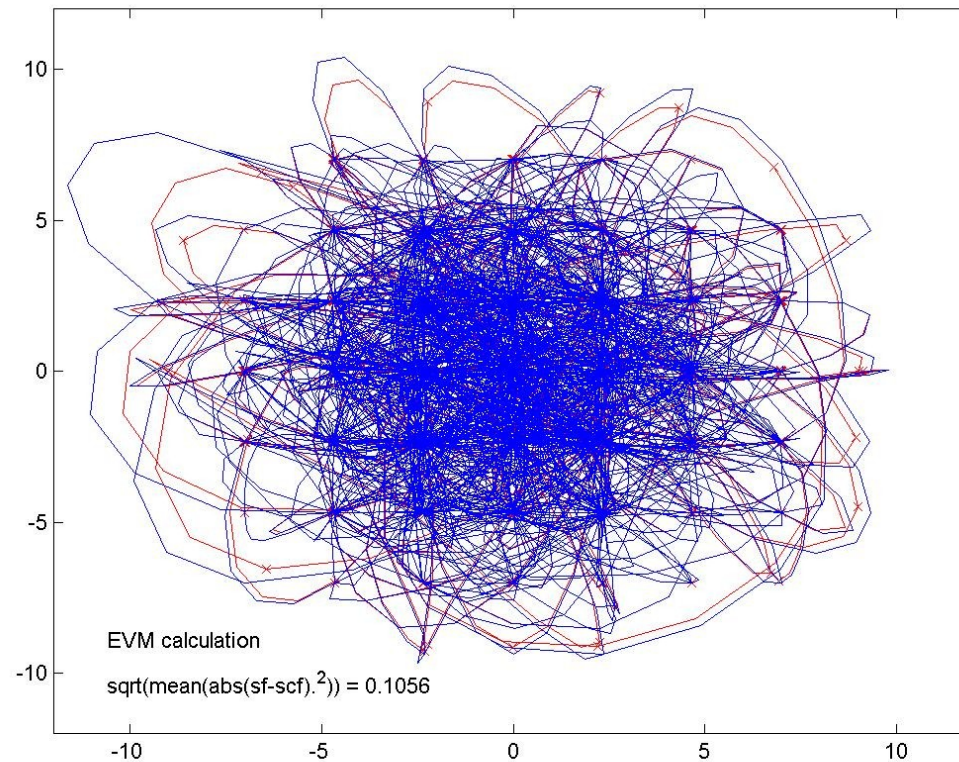


8 Channel W-CDMA signal with 5 x Over sampling

Hard Clipping to 6 dB



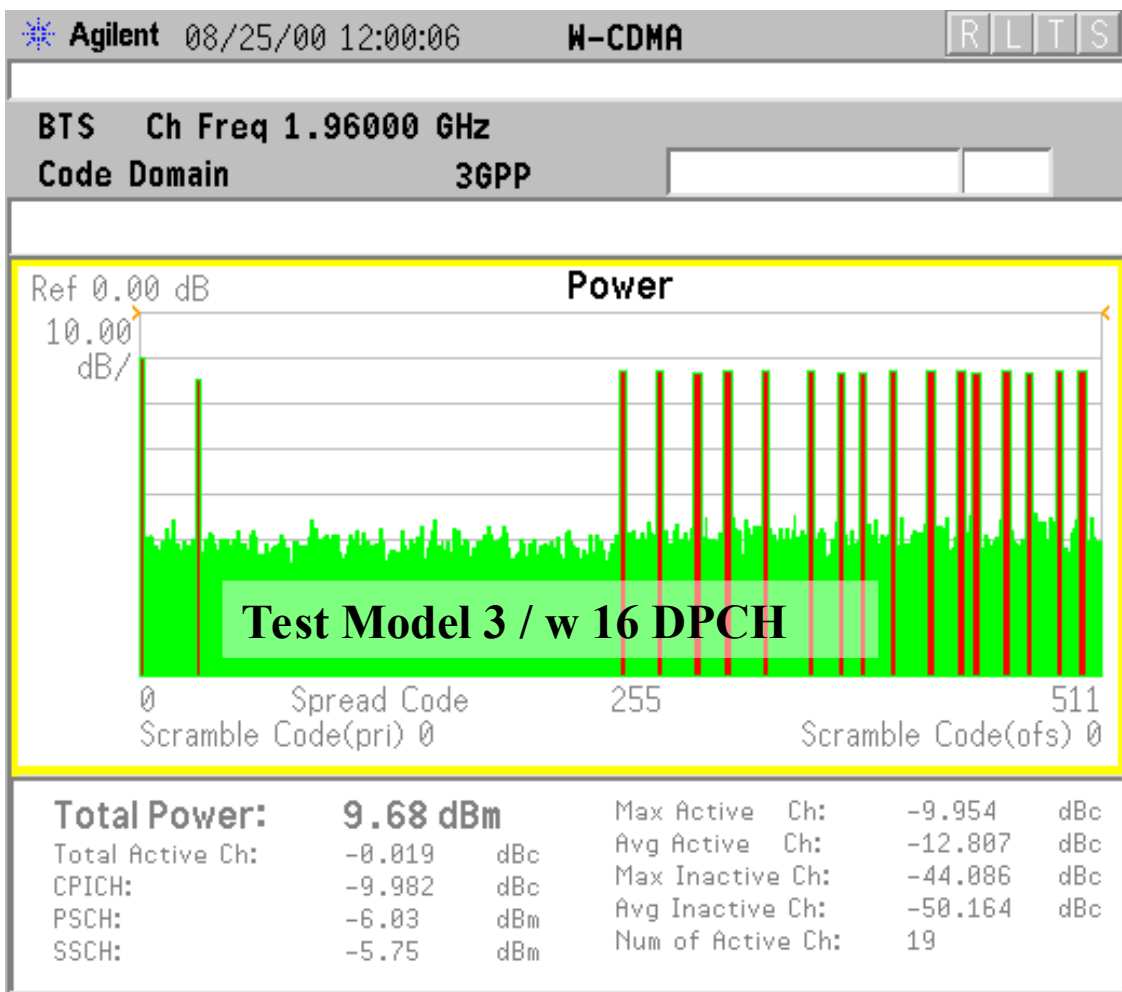
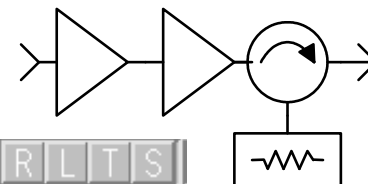
Resulting Errors



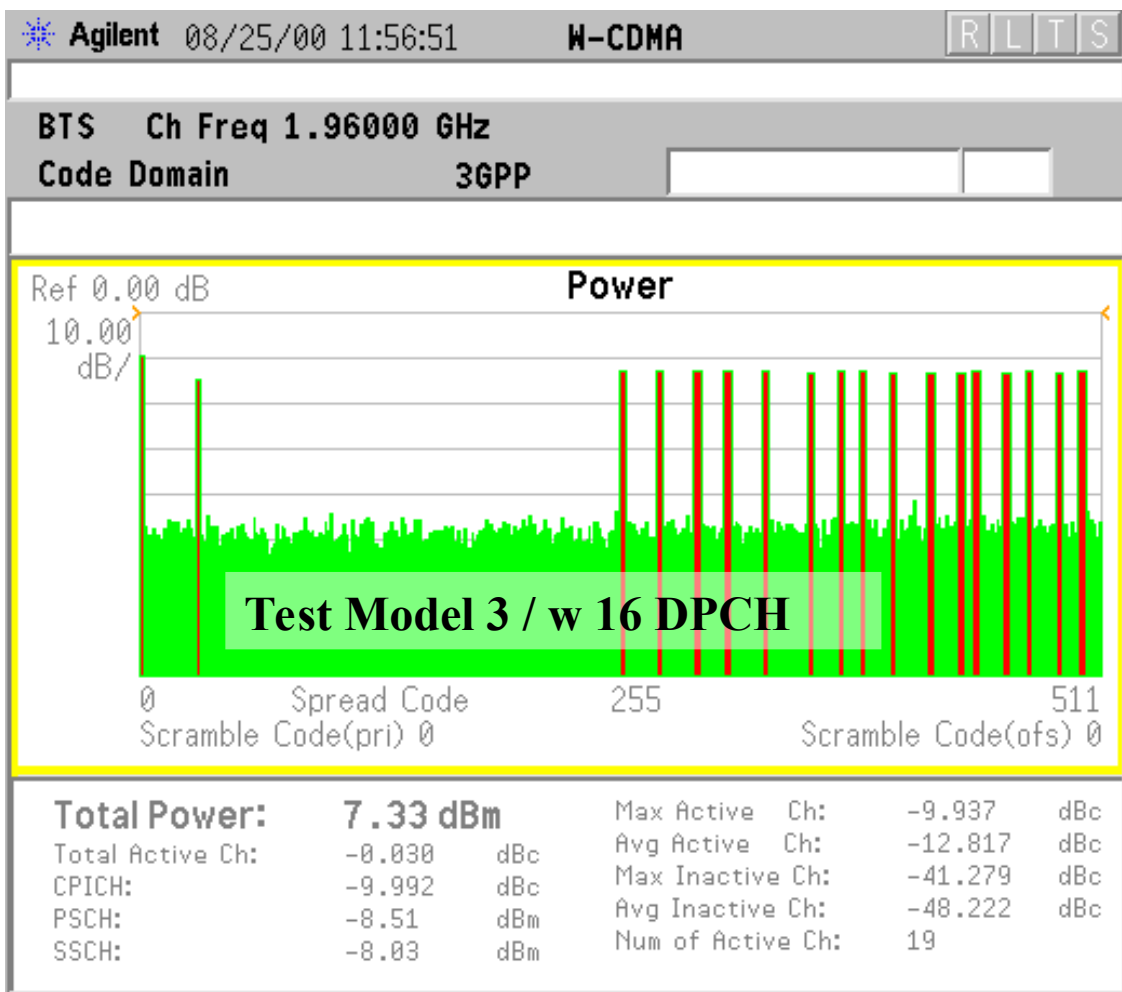
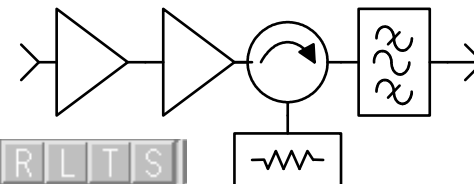
Code Domain Measurement

- Very similar to ρ factor for IS-95
- De-correlate each signal from the composite using its code.
 - Power in each code (channel)
- Un-correlated power is noise
 - De-correlate using all the other codes
 - Unwanted power in unused codes

Code Domain Error



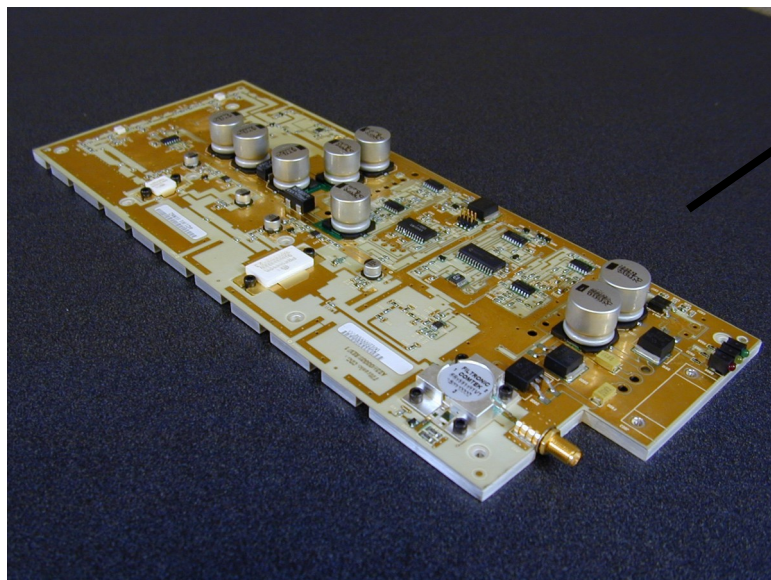
Code Domain Error



Concept Demonstration Unit

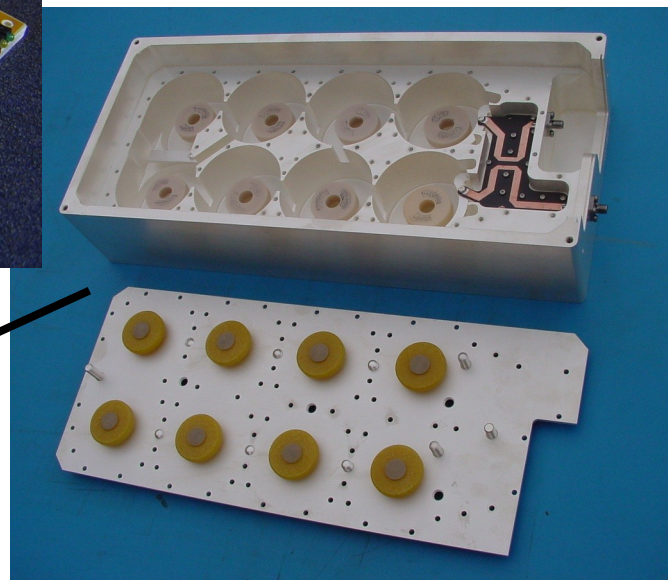
- Single W-CDMA transmit channel
- 40 – 50 dB gain, adjustable
- 30 Watts average
- <6% rms EVM
- <-38dB Code Domain Error
- >15% Efficiency DC in to RF out.

Amplifier and Filter



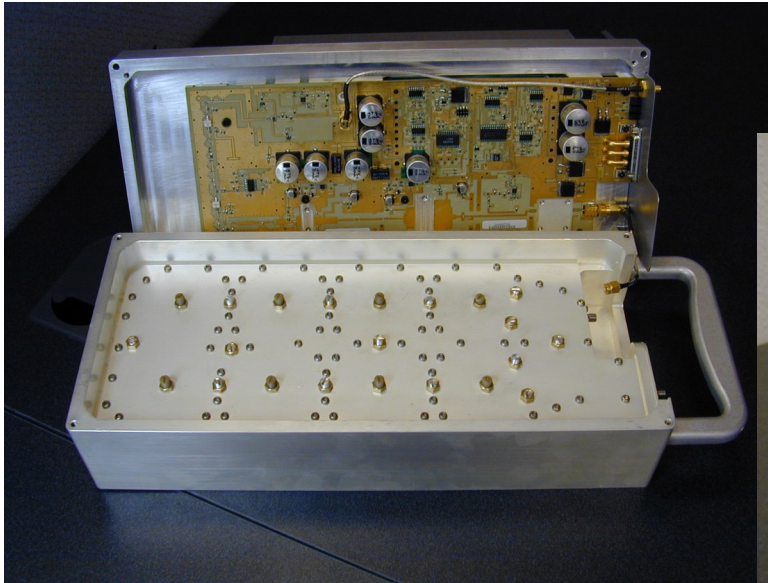
37 watts av.

- 1.2 dB loss
- $\pm 8^\circ$ Pk phase error

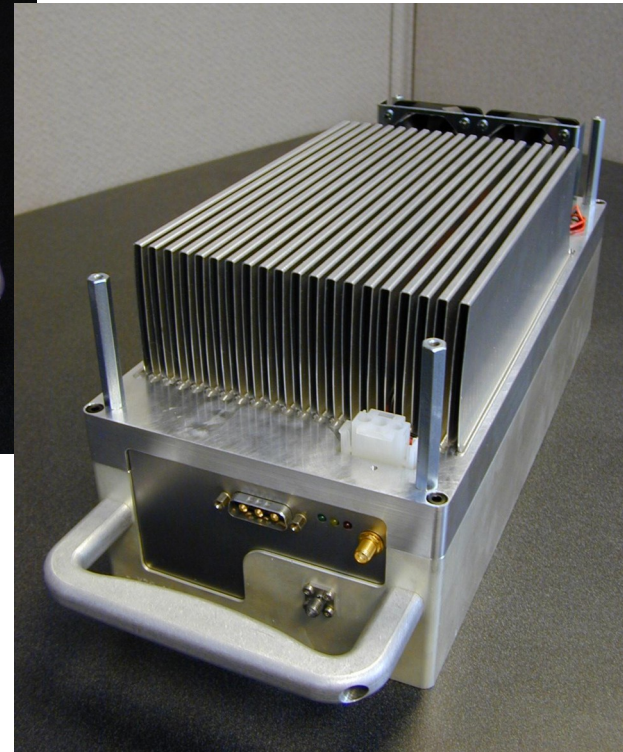


Demonstration Unit

- 28 watts av.
- 15.7% PAE



- ACPR <53 dBc @ 5MHz
- <8% rms EVM
- <-35 dB pk CDE



Filtered Power Amplifier

- Double the output power for same spurious performance
 - 1/2 semiconductor cost
 - Double PAE
- Simple hardware
 - Not error amps, loop control etc
- Single channel only
 - Enough for many sites
 - Duplex and triplex of channels possible

Conclusions

- 2G – 3G has specification differences
 - Rigid frequ. plan in 5 MHz channels
 - Occupied bandwidth / Channel spacing
 - Tough EVM and CDE requirements
- Different compromises are possible and maybe desirable.
- Filtered PA is just one way to re-think the problem.

Acknowledgements

This work represents the combination of work from two groups and many individuals:

Filtronic plc UK R&D Group

Filtronic Comtek Chicago Design Center